

Junmo Lee

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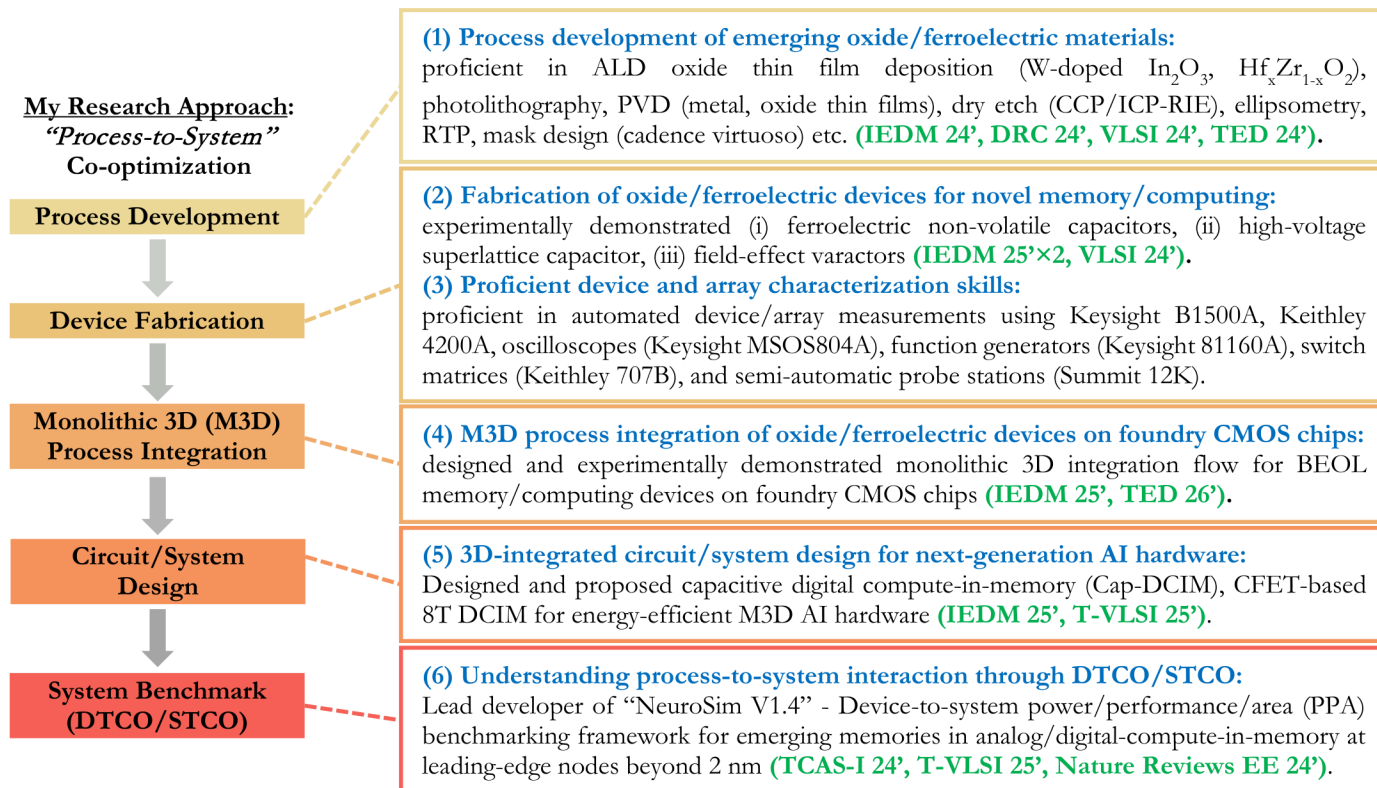
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Google Scholar: <https://scholar.google.com/citations?user=2xu0eTYAAAAJ&hl=en&oi=sra>

SUMMARY

I am an **end-to-end semiconductor device researcher** bridging process development and system-level innovations through **process-to-system co-optimization approach**. I have led cross-layer device research spanning **(i)** process development for **hafnia-based ferroelectric ($\text{Hf}_x\text{Zr}_{1-x}\text{O}_2$)** and **oxide semiconductor (W-doped In_2O_3)** devices; **(ii)** development of their applications in **next-generation computing hardware (e.g., compute-in-memory, AI accelerators)** and **novel memory technologies (e.g., capacitive 3D DRAM, embedded non-volatile capacitive memory)**; and **(iii)** understanding device-to-system interactions through **end-to-end DTCO/STCO framework development (NeuroSim V1.4)**. The overview below summarizes my research experience and skills **((1)-(6))** from process to system, supported by my **first-author publications**.



→ With my process-to-system perspective built on strong device fundamentals, I aim to connect foundry technology development with system design to advance next-generation memory and computing.

EDUCATION

Georgia Institute of Technology (Georgia Tech)

Atlanta, Georgia

Doctor of Philosophy

August 2022 – August 2027 (expected)

Major in Electrical and Computer Engineering, Minor in Physics (currently 4th year Ph.D. Student)

SRC research scholar (Aug 2022-)

Advisor: Dr. Shimeng Yu

Cumulative GPA: 3.9/4.0

Seoul National University (SNU)

Seoul, South Korea

Bachelor of Science

March 2016 - August 2022

Major in Electrical and Computer Engineering

Cumulative GPA: 3.95/4.3 (overall) 3.95/4.3 (major), graduated summa cum laude

HONORS & AWARDS

(1) Award Nomination (2025): Roger A. Haken Best Student Paper Award at IEDM 25'

(2) Award Nomination (2024): Best Oral Presentation Award at DRC 24' (top 15 finalist paper)

(3) Highlight paper (2025): IEDM 25' work on “Monolithic 3D Integration of Dual-Gated ALD Oxide-Channel Non-Volatile Capacitive Memory on 40nm Si CMOS for Digital Compute-in-Memory” selected as conference highlight

(4) **Media Coverage (2025):** IEDM 25' work featured in Nature Electronics (<https://www.nature.com/articles/s41928-025-01533-9>) and Semiconductor Engineering (<https://semiengineering.com/oxides-bring-low-leakage-transistors-to-leading-edge-memories/>)

(5) **Gold Prize in IT Creativity Challenge (2017):** ranked 2nd in undergraduate student circuit design contest held by IEIE (The Institute of Electronics and Information Engineers)

(6) **Excellent Tutor Award (2021):** awarded to excellent peer tutors of the semester at Seoul National University

SKILLS

(1) **Languages:** English (professional working proficiency), Korean (native)

(2) **Microfabrication:** ALD (Zr-doped Hafnia, W-doped Indium Oxide), Photolithography, PVD (metal, oxide thin films), Dry etch (CCP/ICP-RIE), Mask design tools (Cadence Virtuoso, KLayout), Ellipsometry, RTP (rapid thermal processing/annealing)

(3) **Device simulation:** TCAD (Sentaurus, Silvaco)

(4) **Device characterization:** Proficient with performing & scripting electrical characterization using Keysight B1500A, Keithley 4200A, oscilloscopes, function generators, switch matrices, and semi-automatic probe stations (Summit 12K).

(5) **Circuit simulation:** Cadence Virtuoso ADE Explorer

(6) **Programming:** Python, C++, MATLAB

(7) **Code development:** Lead developer of NeuroSim V1.4 - C++/Python-based chip-level benchmarking framework for emerging memory technologies at leading nodes (beyond 3 nm), now widely adopted by the device/circuit research community.

WORK EXPERIENCE

Intel Corporation, Logic Technology Development (LTD)

Incoming Device Engineer/Data Science Intern

Hillsboro, Oregon

August 2026 – November 2026

- Will experience how semiconductor industry manages entire fabrication processes through statistical process control with various engineering analysis tools and techniques.

SMDL (Advisor: Prof. Jong-Ho Lee @ Seoul National University, ECE)

Research Intern

Seoul, South Korea

January 2021 – August 2022

- Optimized sensing-material thickness and post-deposition annealing conditions for resistive and TFT-based gas sensors through device and process analysis as part of a senior design project.
- Developed a hardware-efficient weight-update scheme and a metric for evaluating update noise in synaptic-device-based compute-in-memory systems with abrupt conductance modulation.

NMDL (Advisor: Prof. Sangbum Kim @ Seoul National University, ECE)

Research Intern

Seoul, South Korea

January 2020 – January 2021

- Analyzed training dynamics in synaptic-device-based compute-in-memory systems with nonlinear conductance modulation and developed the Reverse Update Scheme to mitigate the resulting nonlinearity.

Army Missile Command

Republic of Korea Army

Taejeon, South Korea

May 2018 – January 2020

- Earned the Excellence Award (ranked first) in Encryption Contest
- Discharged with army soldier award.

PUBLICATION SUMMARY

** **Refer to PUBLICATIONS** below for the referred papers ([1] [2].. etc.) associated with each project

(1) Device fabrication and characterization of functional metal oxide thin-film capacitors

- Pioneer of ferroelectric non-volatile capacitive memories with ALD oxide channel (IEDM 24' [3], IEDM 25' [1])
- Experimental demonstration of charge-domain in memory search using ferroelectric non-volatile capacitive array (IEDM 25' [2])
- First experimental demonstration of monolithic 3D BEOL integration of ferroelectric non-volatile capacitive memories on TSMC's 40 nm chip (IEDM 25' [1], TED 26' [4])
- Developed BEOL-compatible high-voltage superlattice capacitors and performed comprehensive reliability/electrical characterization, such as time dependent dielectric breakdown (TDDB) characterization (IEDM 23' [5], DRC 24' [11], TED 24' [9]).
- First experimental demonstration of BEOL-compatible DC-DC converters with oxide thin-film transistors and high-voltage superlattice capacitors (VLSI 24' [6])

(2) System-level modeling and benchmarking of emerging device technologies

- Developed NeuroSim V1.4: System-level power/performance/area (PPA) benchmarking framework for emerging device and compute-in-memory technologies at advanced technology nodes (TCAS-I 24' [7], T-VLSI 25' [8], Nature Reviews EE 24' [10], DAC 24' [15])
- Performed DTCO of Digital Compute-in-Memory at leading-edge technology nodes down to 0.5 nm technology node (TCAS-I 24' [7], T-VLSI 25' [8])

(3) Device-to-system codesign of AI hardware technologies

- Designed a novel AI accelerator (Capacitive Digital-Compute-in-Memory) based on experimentally demonstrated non-volatile capacitive memories on TSMC's 40nm chip (IEDM 25' [1])
- Designed a novel 8T Digital Compute-in-Memory based on 0.5 nm CFET technology (T-VLSI 25' [8], APCCAS 24' [13])
- Designed a novel weight update algorithm for AI accelerators based on analog synaptic devices (JXCDC 22' [14])

PUBLICATIONS

* indicates equal contribution

- [1] [Selected Publication] J. Lee, C. Zhang, S. Park, H. Park, T.-H. Kim, L. Liu, E. Ambrosi, M. Song, X. Bao, M.-F. Chang, S. Datta, S. Yu, "Monolithic 3D Integration of Dual-Gated ALD Oxide-Channel Non-Volatile Capacitive Memory on 40nm Si CMOS for Digital Compute-in-Memory," **IEEE International Electron Devices Meeting (IEDM) 2025**, San Francisco, USA.
- (highlight paper, nominated for the best student paper award)
- [2] [Selected Publication] O. Phadke*, J. Lee*, P.-K. Hsu, C. Zhang, A. Khan, S. Datta, S. Yu, "Array-Level Demonstration of Charge-Domain In Memory Search using Back-End-of-Line Compatible Ferroelectric Nonvolatile Capacitors," **IEEE International Electron Devices Meeting (IEDM) 2025**, San Francisco, USA.
- [3] [Selected Publication] J. Lee, C. Zhang, M. Shon, J. Read, S. Deng, O. Phadke, P. V. Ravindran, M. Tian, Y.-C. Luo, T.-H. Kim, A. I. Khan, S. Datta, S. Yu, "BEOL-compatible non-volatile capacitive synapse with ALD W-doped In₂O₃ semiconductor layer," **IEEE International Electron Devices Meeting (IEDM) 2024**, San Francisco, USA.
- [4] [Selected Publication] J. Lee, C. Zhang, S. Park, H. Park, T.-H. Kim, L. Liu, E. Ambrosi, M. Song, X. Bao, M.-F. Chang, S. Datta, S. Yu, "Dual-Gated ALD Oxide-Channel Non-Volatile Capacitive Memory Monolithically Integrated on Foundry 40 nm CMOS for Capacitive Digital Compute-in-Memory," **IEEE Transactions on Electron Devices**, vol. 73, no. 5, pp. 3091-3101, 2026.
- (invited paper as a top-ranked student paper at IEDM 2025)
- [5] [Selected Publication] S. Deng, J. Kwak, J. Lee, K. A. Aabrar, T.-H. Kim, S. Kirtania, G. Choe, C. Zhang, W. Li, O. Phadke, S. Yu, S. Datta, "BEOL compatible oxide power transistors for on-chip voltage conversion in heterogenous 3D (H3D) integrated circuits," **IEEE International Electron Devices Meeting (IEDM) 2023**, San Francisco, USA.
- [6] [Selected Publication] S. Deng*, J. Kwak*, J. Lee*, D. Chakraborty, J. Shin, O. Phadke, S. G. Kirtania, C. Zhang, K. A. Aabrar, S. Yu, S. Datta, "Demonstration of on-chip switched-capacitor DC-DC converters using BEOL compatible oxide power transistors and superlattice MIM capacitors," **IEEE Symposium on VLSI Technology and Circuits (VLSI) 2024**, Hawaii, USA.
- [7] [Selected Publication] J. Lee, A. Lu, W. Li, S. Yu, "NeuroSim V1.4: Extending technology support for digital compute-in-memory towards 1nm node," **IEEE Transactions on Circuits and Systems I: Regular Papers**, vol. 71, no. 4, pp. 1733-1744, 2024.
- [8] [Selected Publication] J. Lee, M. Shon, F. Waqar, S. Yu, "3-D Digital Compute-in-Memory Benchmark With A5 CFET Technology: An Extension to Lookup-Table-Based Design," **IEEE Transactions on Very Large Scale Integration Systems**, 2025, accepted.
- [9] [Selected Publication] S. Deng*, J. Kwak*, J. Lee*, D. Chakraborty, J. Shin, O. Phadke, S. G. Kirtania, C. Zhang, K. A. Aabrar, S. Yu, S. Datta, "Monolithic switched-capacitor DC-DC voltage converters using BEOL-compatible oxide power transistors and superlattice MIM capacitors," **IEEE Transactions on Electron Devices**, vol. 71, no. 12, pp. 7999-8006, 2024.
- [10] [Selected Publication] Anni Lu*, J. Lee*, T.-H. Kim*, M. Karim, R. Park, H. Simka, S. Yu, "High-speed emerging memories for AI hardware accelerators," **Nature Reviews Electrical Engineering**, vol. 1, pp. 24-34, 2024.
- [11] [Selected Publication] J. Lee, S. Deng, J. Kwak, M. Shon, S. Datta, S. Yu, "Optimization of backside of silicon-compatible high voltage superlattice capacitor for 12V-to-6V on-chip voltage conversion," **IEEE Device Research Conference (DRC) 2024**, College Park, MD. (Nominated for the best oral presentation award)
- [12] S. Deng, H.-C. Chou, C. Zhang, J. Kwak, A. D. Asokan, E. Quezada, E. Sarkar, J. Lee, J. Shin, D. Chakraborty, V. Sathe, S. Yu, S. Datta, "High-efficiency BEOL-compatible on-chip switched capacitor DC-DC converter using planar and vertical oxide transistors with trench capacitors," **IEEE International Electron Devices Meeting (IEDM) 2025**, San Francisco, USA.
- [13] M. Shon*, J. Lee*, S. Yu, "3D digital compute-in-memory benchmark with A5 CFET technology," **IEEE Asia Pacific Conference on Circuits and Systems (APCCAS) 2024**, Taipei, Taiwan.
- [14] J. Lee, J. Hwang, Y. Cho, M. K. Park, W. Y. Choi, S. Kim, J. H. Lee, "CRUS: A Hardware-Efficient Algorithm Mitigating Highly Nonlinear Weight Update in CIM Crossbar Arrays for Artificial Neural Networks," **IEEE Journal on Exploratory Solid-State Computational Devices and Circuits**, 8(2), 145-154, 2022.
- [15] A. Lu, J. Lee, Y.-C. Luo, H. Li, I. Young, S. Yu, "Digital CIM with noisy SRAM bit: A compact clustered annealer for large-scale combinatorial optimization," **ACM/IEEE Design Automation Conference (DAC) 2024**, San Francisco, CA.
- [16] S. Deng, J. Kwak, J. Lee, S. Yu, S. Datta, "BEOL-compatible on-chip DC-DC converters," **IEEE International Interconnect Technology Conference (IITC) 2024**, San Jose, CA.

- [17] G. Jeong, J. Kwak, M. Vadlamani, **J. Lee**, S. Yu, "Physics-Informed Neural Network-Assisted Compact Modeling for Process Variation in Non-Volatile Capacitive Crossbar Arrays," **IEEE International Reliability Physics Symposium (IRPS) 2026**, Tucson, AZ, USA, pp. 1–4.
- [18] D. Yu, H. Hwang, B. Oh, **J. Lee**, S. Yu, T.-H. Kim, H. Kim, "A Unified Flash Memory Platform for Mode-Adaptive and Robust AI Computation," **Advanced Science**, e76619, 2026.
- [19] J. Kwak, G. Choe, **J. Lee**, S. Yu, "Monolithic 3D transposable 3T embedded DRAM with back-end-of-line oxide channel transistor," **IEEE International Symposium on Circuits and Systems (ISCAS) 2024**, Singapore.
- [20] J. Kwak, S. Deng, **J. Lee**, S. Datta, S. Yu, "Backside active power delivery with hybrid DC-DC converter enabled by amorphous oxide semiconductor transistors," **IEEE Transactions on VLSI Systems**, 2025.
- [21] J. Kwak, S. Deng, C. Zhang, G. Jeong, O. Phadke, S.G. Kirtania, **J. Lee**, S. Datta, S. Yu, "Machine Learning-Assisted Compact Modeling of AC Stress-Induced Bias Temperature Instability in Top- and Bottom-Gate W-Doped In₂O₃ Channel Semiconductor Transistors," **IEEE Transactions on Electron Devices**, 2025.
- [22] O. Phadke, P.-K. Hsu, Y.-C. Luo, J. Sharda, V. Garg, J. Read, **J. Lee**, H. Mulaosmanovic, S. Dunkel, S. Beyer, S. Yu, "Ternary content addressable memory with ferroelectric non-volatile capacitor," **IEEE Electron Device Letters** 2025.
- [23] J. Kwak, C. Zhang, G. Jeong, O. Phadke, S. G. Kirtania, **J. Lee**, S. Datta, S. Yu, "Machine learning-assisted modeling of AC stress-induced bias temperature instability in oxide channel transistors for 2T gain cell eDRAM," **IEEE International Reliability Physics Symposium (IRPS) 2025**, Monterey, CA.
- [24] M. Vadlamani, J. Jia, T. Xie, Y.-C. Luo, **J. Lee**, S. Li, S. Yu, "Capacitive Crossbar Array for Solving Matrix Equations in One-Shot," **IEEE Electron Device Letters**, vol. 46, no. 3, pp. 389-392, 2025.
- [25] W. Li, **J. Lee**, S. Yu, "Optimization strategies for digital compute-in-memory from comparative analysis with systolic array," **IEEE International Conference on Artificial Intelligence Circuits & Systems (AICAS) 2023**, Hangzhou, China.
- [26] W. Li, X. Zhang, **J. Lee**, F. L. Degertekin, S. Li, S. Yu, "Enabling ultra-low power ultrasound imaging with compute-in-memory sparse reconstruction accelerator," **IEEE Biomedical Circuits and Systems Conference (Bio-CAS) 2023**, Toronto, Canada.
- [27] W. Shin, B. Choi, **J. Lee**, S. Hong, Y. Jeong, G. Jung, J. Park, D. Kim, C. Lee, J. H. Lee, "Sub-ppm Detection of NO₂ gas using IGZO TFT-type Gas Sensor with Optimal Signal-to-Noise Ratio", **International Conference on Solid State Devices and Materials (SSDM) 2021**, virtual.

Under Review

- [28] **J. Lee**, C. Zhang, S. Datta, S. Yu, "Pulse-Mode Operation and Reliability of BEOL-Compatible Ferroelectric Non-Volatile Capacitive Memories with Amorphous Oxide Semiconductor Channels," **IEEE Transactions on Nanotechnology** (journal extension paper of IEDM 2024)
- [29] **J. Lee**, K. Lee, C. Zhang, T.-H. Kim, M. Tian, L. J.-C. Liu, M.-F. Chang, S. Datta, S. Yu, "Logic-Compatible Capacitive Gain Cell Monolithically Integrated on Si CMOS for Bit-Cost-Scalable 3D Digital Compute-in-Memory," submitted to IEEE International Electron Devices Meeting (IEDM) 2026.

TEACHING EXPERIENCE

Undergraduate Tutor (@ Seoul National University)

Peer Tutor (Semiconductor Devices)

Seoul, South Korea

September 2021 – December 2021

- Provided 30 hours of online lecture, received Excellent Tutor Award

Undergraduate Tutor (@ Seoul National University)

Peer Tutor (Introduction to Electronic Circuits and Laboratory)

Seoul, South Korea

September 2022 – December 2022

- Provided 24 hours of online lecture

SCHOLARSHIPS

(1) Merit-based Scholarship, Seoul National University (2020): provided to undergraduate students with excellent academic record

(2) Scholarship, Korea Electric Power Corporation (2021): provided to undergraduate students with excellent academic record